

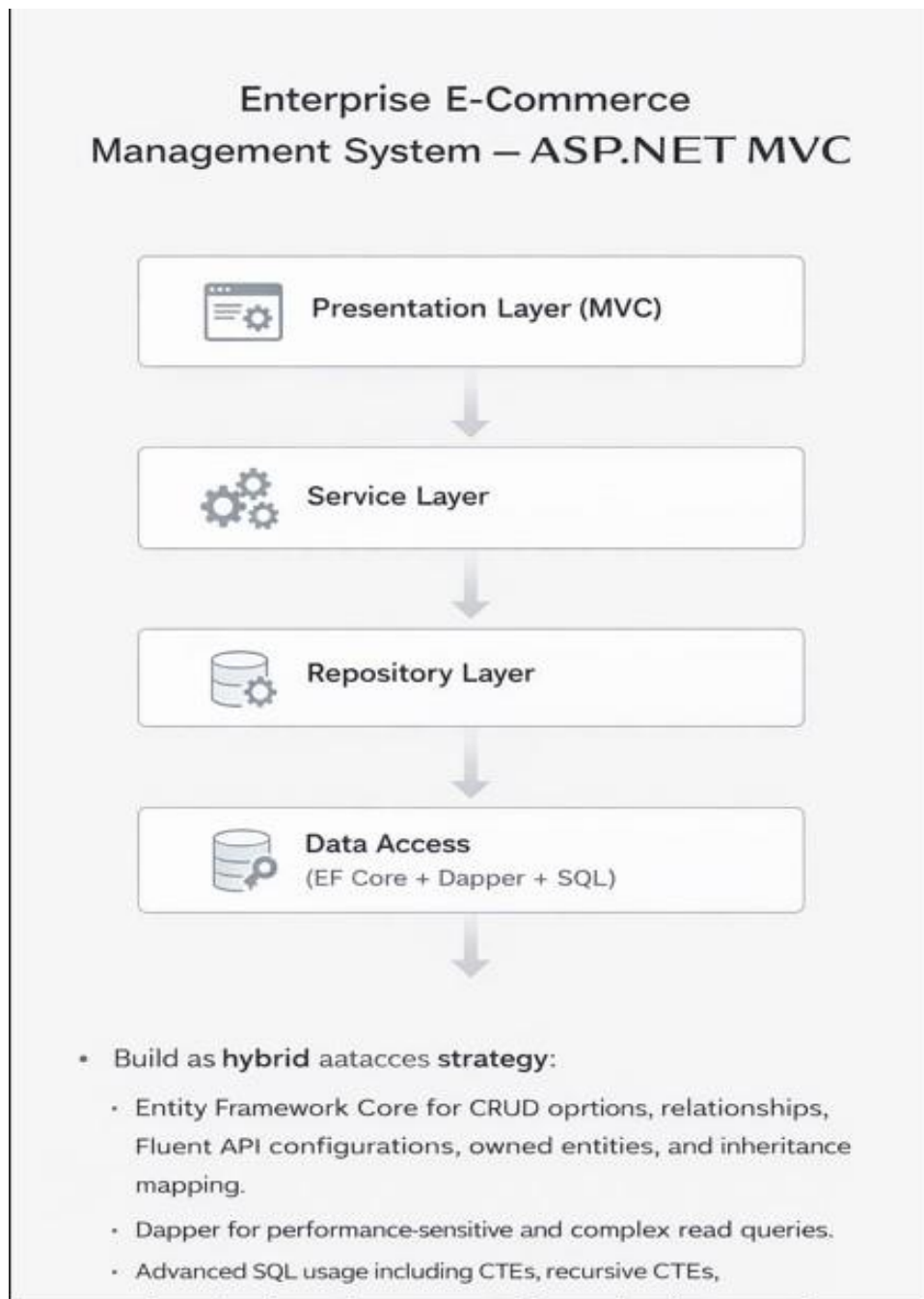
Enterprise E-Commerce Management System

Backend Architecture Case Study

Designed & Implemented using ASP.NET MVC, EF Core, and Dapper

This project demonstrates a scalable, maintainable, and high-performance backend system suitable for real-world applications.

System Architecture



The system follows a clean layered architecture ensuring separation of concerns and maintainability.

Layers:

- Presentation Layer (MVC): Handles UI and request routing.
- Service Layer: Business logic and workflows.
- Repository Layer: Data abstraction.
- Data Access: EF Core + Dapper hybrid strategy.

Technical Decisions

A hybrid data access strategy is used:

- EF Core for CRUD, relationships, and maintainability.
- Dapper for performance-critical queries.

Advanced SQL usage:

- CTEs and recursive queries.
- Optimized joins and indexing strategies.

Performance & Scalability

- Server-side pagination for large datasets.
- Optimized queries using Dapper.
- Clean architecture enables scaling and feature expansion.

Conclusion

This backend system is designed to be production-ready, scalable, and easy to maintain. It reflects real-world backend engineering practices.